Experiment 3: when does a push end, and why is a ramp not symmetric?

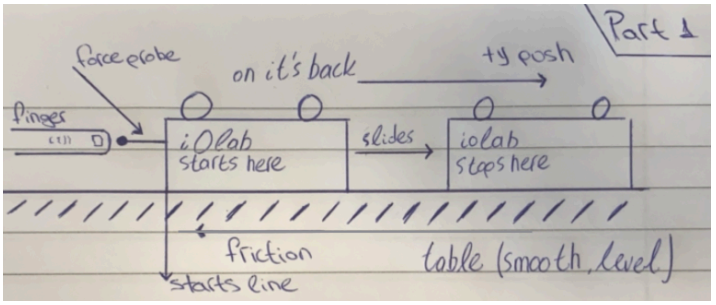
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Introduction:

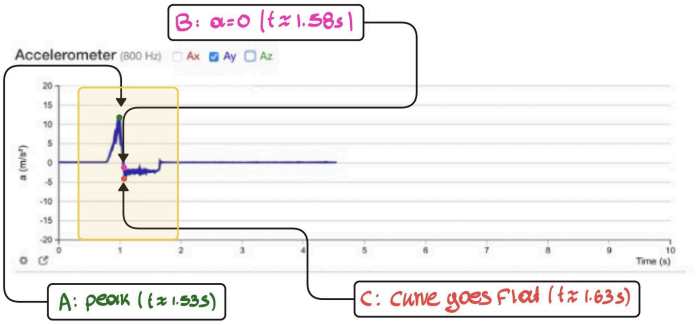
Conveyor belts in logistics and manufacturing industries need to move effectively. One of the things that makes sure of that is the timing of packages moving from one platform to another. We want to experiment in order to figure out how friction and acceleration plays part in an object's behaviour on a horizontal and at an angle. This will help the programming and prediction for the whole operation, so that it later can be automated.

Part 1: when does the hand lose contact?

As an object moves along the conveyor belt, it might need to stop to insure safe transportation of other objects, without crashing. With this we naturally need to be able to predict the acceleration and deacceleration, so that the behavior later can be implemented in the automation.



Sketch of operations for horizontal experiment, iOLab device on it's back

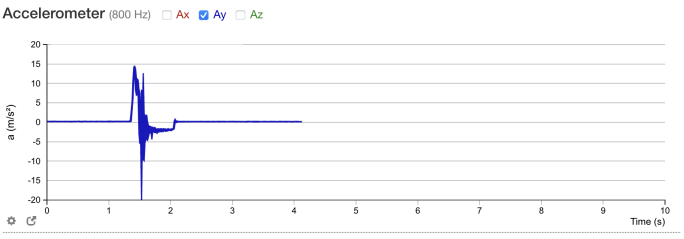


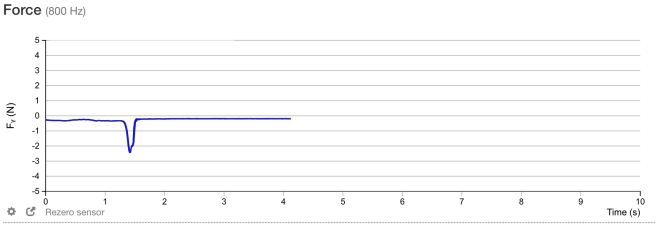
Initial test result

The iOLab was placed on its back on a horizontal surface, with the wheels facing upward. The device was pushed along the y-axis and allowed to slide until it stopped. Several trials were performed to investigate when the hand lost contact with the device. Acceleration data were recorded using the iOLab's accelerometer.

After the initial test result we started debating which point would be the loss of contact between the hand and the iOLab, we decided that point C is most likely. Thus we started making more behavioral experiments to prove the hypothesis true. The tests consisted of five measurements where we pushed and let go of the accelerometer, one where we never let go but stopped it ourselves and one where we pulled just to emphasise that pulling would only mirror the result and not completely change them. The result of these tests can be found on the table below.

To check this in a second way, we recorded one run with the iOLabs force measuring ability and pushed on the end with the sensor itself. The sensor measures the force from the hand directly, so it shows when the push ends.





In the figure above, the acceleration changes in response to the applied force. When the force returns to approximately zero, indicating that the hand has lost contact with the iOLab, the acceleration settles into a flatter region. This observation supports Hypothesis C, which states that the hand loses contact when the acceleration flattens out.

Theory:

According to Newton's Second Law, the net force acting on an object is equal to its mass multiplied by its acceleration: $\Sigma F = ma$. When the iOLab is pushed horizontally, the applied force from the hand and the friction from the surface determine its acceleration. Once the hand loses contact, the applied force becomes zero and the friction is the only horizontal force

acting on the device. While the iOLab slides, $m \cdot a = F_{hand} - \mu k \cdot m \cdot g$, so $F_{hand} = m \cdot (a - a_f)$ with $a_f = -\mu k \cdot g$ and $\mu k = -a_f/g$ ($g = 9.82$ m/s²). At B the acceleration is zero, so $F_{hand} = \mu k \cdot m \cdot g$ and the hand still pushes. Only at C, where $a = a_f$, is $F_{hand} = 0$. Hypothesis: the hand loses contact at C. We compare a at each point with a_f using $z = |x_1 - x_2| / \sqrt{(\delta x_1^2 + \delta x_2^2)}$. A point is rejected if $z > 3$ and not rejected if $z \leq 2$.

Table 1: Part 1, five runs, accelerometer at 800 Hz. Times are ± 0.010 s (one pixel). A is the peak, B the first zero after A, and C the first point after B inside $a_f \pm 2s$, where s is the scatter of the graph on the flat part. t_{stop} is when the iOLab stops. z_A compares a_A with a_f, and z_B compares 0 with a_f.

Run	t_A (s)	t_B (s)	t_C (s)	t_{stop} (s)	a_A (m/s²)	a_f (m/s²)	μ_k	z_A	z_B
1	0.900	1.104	1.132	1.658	4.42 ± 0.14	−2.41 ± 0.12	0.245 ± 0.012	42	20
2	1.737	1.860	1.875	2.496	5.6 ± 0.4	−2.63 ± 0.12	0.267 ± 0.012	21	22
3	0.461	0.512	0.548	1.000	6.1 ± 0.2	−2.18 ± 0.12	0.222 ± 0.012	32	18
4	1.215	1.261	1.277	1.903	6.8 ± 0.5	−2.82 ± 0.14	0.287 ± 0.014	19	20
5	1.697	1.806	1.816	2.446	6.9 ± 0.3	−2.44 ± 0.12	0.248 ± 0.012	31	20
Mean	-	-	-	-	6.0 ± 0.5	−2.49 ± 0.11	0.254 ± 0.011	18	23

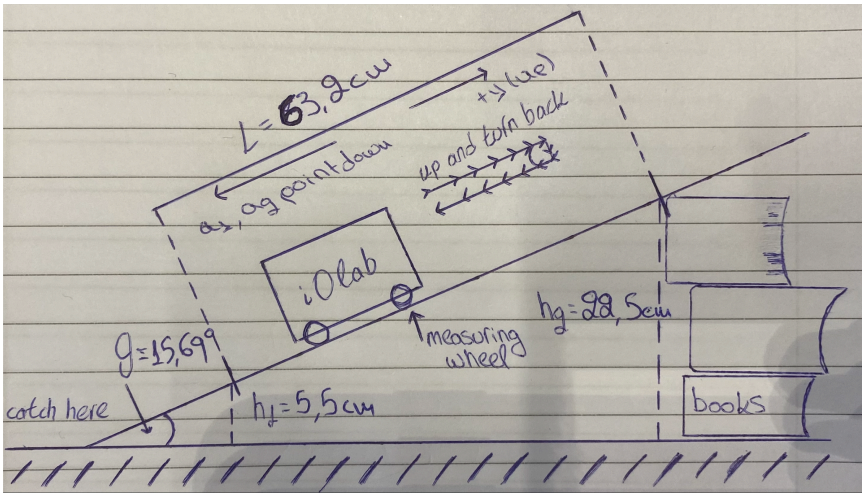
Over the five runs, $a_f = (-2.49 \pm 0.11)$ m/s², so $\mu_k = 0.254 \pm 0.011$. At the hand force per mass, $a - a_f$, is (8.5 ± 0.5) N/kg with $z_A = 18$. At B it is (2.49 ± 0.11) N/kg with $z_B = 23$. So at the moment of top speed the hand is still pushing as hard as friction, and A and B are rejected in every run. At C the acceleration reaches a_f and the hand force is zero, which fits the force run in Figure 3. Hypothesis C is not rejected. The drop from 0 at B to a_f at C takes (0.021 ± 0.007) s, only about two pixels on the graph.

Part 2: rolling up and down a ramp

Objective:
The aim of this experiment is to investigate the motion of the iOLab as it rolls up and down an inclined plane. We compare its acceleration while moving uphill and downhill to understand the effects of gravity and friction.

Theory:
When the device rolls up the inclined plane, gravity acts against the motion, causing it to slow down. When it rolls back down, gravity acts in the direction of its motion, causing it to accelerate. The gravitational force acting on the iOLab is given by: $F_g = mg$. The component of gravity parallel to the inclined plane is: $F_{||} = mg\sin\theta$, where m is the mass of the device, g is the gravitational acceleration and θ is the angle of the incline as shown on the sketch.
According to Newton's Second Law, the net force acting on the device determines its acceleration: $\Sigma F = ma$. Friction and rolling resistance oppose the motion and affect the net force. When the iOLab rolls uphill, gravity and friction act downhill, resulting in: $ma_1 = -(mg\sin\theta + F_f)$. When the iOLab rolls downhill, gravity still points downhill, but the friction now points uphill: $ma2 = -mg \cdot \sin\theta + Ff$.

Hypothesis:
We expect $|a1| > |a2|$. Gravity contributes the same component $g \cdot \sin\theta$ in both directions, but rolling resistance opposes the motion. When the iOLab rolls uphill, rolling resistance acts downhill and adds to gravity. When it rolls downhill, rolling resistance acts uphill and reduces the net acceleration. Test: $D = |a_1| - |a2|$ and $z = D/\delta D$. The hypothesis is supported if $D>0$ and $z>3$. If $z \leq 2$, the difference is not statistically significant, and the hypothesis is not supported by the measurements. A result between these thresholds is inconclusive.



Sketch of operation for angled experiment

A flat board was placed at an incline making the h_2 = 22.5 cm and the angle 15.60°. However we did not start the measurements at the very bottom of the board, we did it in the middle of the board so the test range is 63.2 cm. Five measurements were recorded. A hand was placed at the bottom to catch the iOLab, and another used to push the iOLab up the incline. The recording was ended after it returned to the hand.

Table 2: Part 2, five runs. a_1 and a_2 are the slopes of straight lines fitted to the velocity graph. The up window runs from 0.10 s after top speed to 0.10 s before the turn, and the down window from 0.10 s after the turn to 0.08 s before the catch. D = |a1| − |a2| and z = D/δD. F_f/m from the wheel is D/2. From the accelerometer it is half the difference between the readings rolling down and rolling up.

Run	a_1 (m/s²)	a_2 (m/s²)	D (m/s²)	z	F_r/m, wheel (m/s²)	F_r/m, accelerometer (m/s²)
1	−3.04 ± 0.03	−2.030 ± 0.011	1.01 ± 0.03	32	0.504 ± 0.016	0.41 ± 0.07
2	−2.94 ± 0.02	−1.998 ± 0.017	0.94 ± 0.03	35	0.469 ± 0.013	0.47 ± 0.07
3	−3.23 ± 0.02	−1.79 ± 0.03	1.44 ± 0.04	41	0.720 ± 0.018	0.67 ± 0.07
4	−2.87 ± 0.03	−1.913 ± 0.013	0.95 ± 0.03	31	0.478 ± 0.016	0.33 ± 0.07
5	−2.971 ± 0.019	−1.897 ± 0.019	1.07 ± 0.03	40	0.538 ± 0.014	0.43 ± 0.07
Mean	−3.01 ± 0.06	−1.93 ± 0.04	1.08 ± 0.07	14.5	0.54 ± 0.04	0.46 ± 0.06

D = (1.08 ± 0.07) m/s² gives z = 14.5, and each run on its own has z of at least 31. The hypothesis $|a_1| > |a_2|$ is not rejected: the iOLab slows down faster going up than it speeds up coming down. Half the difference gives F_f/m = (0.54 ± 0.04) m/s², so $\mu_r = F_f/(m \cdot g \cdot \cos\theta)$ = 0.057 ± 0.004. The accelerometer measures the same friction directly, (0.46 ± 0.06) m/s², and the two agree (z = 1.2). Run 3 has the most friction with both sensors.

The mean $(|a_1| + |a_2|)/2$ = (2.47 ± 0.04) m/s² should equal $g \cdot \sin\theta$ = (2.64 ± 0.02) m/s², but it is lower (z = 4.0). At rest the accelerometer reads (2.38 ± 0.05) m/s². That is close to the wheel value (z = 1.4) and far from the ruler (z = 4.6). The sensors measure the slope against gravity, while the ruler measures it against the table, so a table that is not quite level could explain the gap. The spinning wheels also lower the mean a little.

Error sources: we read every value from screenshots of the app graphs, so one pixel limits the precision. In Part 1 that is 0.010 s and 0.16 m/s². Moving the Part 2 windows by 0.05 s changes a1 by up to 0.1 m/s², so we use the spread between runs (SDOM) as the uncertainty of the mean. In Part 1, if the reading of 0.2 m/s² at rest comes from a tilted table and not from the sensor, μk is 0.233 instead of 0.254.

Reflections:

1) **How did you approach today's experiment: did you have expectations of the final result, or did you only get the result after you had the data?**
   We went in with thinking that it was A or B for the first test, after debating and during the first test we realized it would most likely be C, which was proven when we finished the testing. For the second experiment, we had no clear expectations about the results.

2) **What did you learn from today's experiment?**
   We learned how to utilize more of iOLabs functions in order to collect and analyze data. Doing so, made the experiments a lot easier to understand and prove. Furthermore, we discovered that the hand continues to push the device even after it reaches its maximum velocity, which surprised us.

3) **What do you think you were supposed to learn from today's experiment?**
   We believe the purpose of this experiment was to learn how to formulate testable hypotheses and design experiments to support or reject them. We also learned the importance of analyzing our measurements using Newton's Second Law and considering uncertainties to ensure that our conclusions are supported by the data rather than simply trusting the results.